Methods survey - Modern methods

Dimensionality reduction

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Statistical models

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Dynamics/state space models

Mechanistic models

Inferring connectivity

Spatiotemporal pattern analysis

Machine learning models

Comparing data to ML models

Variational inference/ELBO

Learning and inference techniques

Information theoretic methods

Dimensionality reduction (I)

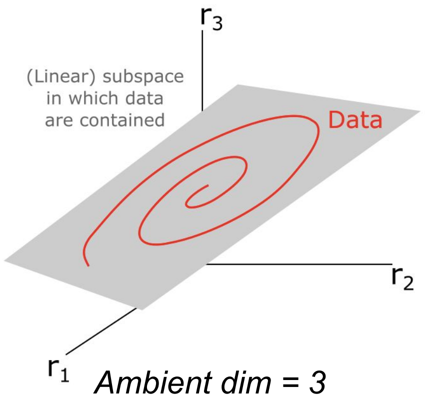
Project or embed data onto "lower-dimensional" subspace or manifold

Often data = population activity patterns, e.g. firing rates estimated by binning spike trains

There are MANY definitions of dimensionality

Three important classes

- Ambient dimensionality: How many numbers per data point in experimental recording
  - Usually determined by experiment setup rather than intrinsic structure in data
- Linear dimensionality: Number of basis vectors needed to span subspace containing data — captures linear structure in data
  - PCA-dimensionality: # of principal components needed to explain X% of total variance
    - Dependent on choice of X
  - Participation ratio: Continuous-valued measure of linear dimensionality
    - Gao et al *bioRxiv* 2017, Recanatesi et al *Patterns* 2022
- Nonlinear dimensionality: How many numbers are needed to specify where a data point lives in the ambient space — captures nonlinear structure in data
  - Many methods to estimate this from data
  - Often called "intrinsic" dimensionality



Ambient dim = 3
Linear dim = 2
Nonlinear dim = 1

Dimensionality reduction (II) - Linear methods

- Principal Components Analysis (PCA)
  - Compute covariance matrix of dataset matrix
  - Perform eigendecomposition
  - Returns list of (orthogonal) eigenvectors sorted by how much variance in the data they explain
  - Project data onto top K eigenvectors to reduce to K dimensions

- Singular Value Decomposition (SVD)
  - Mathematically equivalent to PCA applied to both rows and columns of dataset

- jPCA (Churchland et al, *Nature* 2012)
  - PCA variation for identifying rotational dynamics
  - Requires time-varying data
  - Assumes time derivative is rotational matrix (with purely imaginary eigenvalues) applied to system state

- Canonical Correlation Analysis (CCA)
  - Finds projections of dataset A and dataset B that are maximally correlated
  - Hotelling, *Biometrika* 1936
  - Hardoon et al, *Neur Comput* 2004

Dimensionality reduction (III) - Nonlinear methods

- Multi-dimensional scaling (MDS)
  - Kruskal, Psychometrika 1964
- Papers using MDS in neuroscience:
  - Shepard, *Science* 1987
- Local linear embedding
  - Roweis & Saul, Science 2000
- IsoMAP
  - Tenenbaum et al, Science 2000
- t-SNE
  - van der Maaten & Hinton, JMLR 2008
- Papers using t-SNE
  - Berman et al, *J R Soc Interface* 2014
- UMAP
  - McInnes et al, arXiv 2018
- Gaussian Process Factor Analysis
  - Yu et al, *J. Neurophys.* 2009
- LFADS
  - Pandinarath et al *Nature Methods* 2018
- Papers using LFADS:
  - Ames & Churchland, *eLife* 2019
- Manifold inference of neural dynamics (MIND)
  - Nieh, Schottdorf et al, *Nature* 2021
- CEBRA
  - Schneider et al *Nature Methods* 2023

Clustering and segmentation

Goal:

- Split time-series into underlying “states”, each corresponding to one time of pattern
  - E.g. an animal locomotion signal might be split into “walking”, “still”
- *True* number of states is usually not very well defined
  - Scientist must usually pick a number or resolution of states, often controlled by model parameter
- Can be done with simple methods such as threshold crossing (e.g. “walking” → velocity > v_{min}
- These days more common to identify “hidden states” via statistical model
  - Capture variability of observations via generative model
  - Well-posed to incorporate extensions, “stickiness” of states, priors over state distributions, etc.

<u>Example papers where clustering/segmentation is used to process neural and/or behavior time-series</u>

Kemere et al., Journal of Neurophysiology, 2008, https://doi.org/10.1152/jn.01280.2007
Berman et al., Nature Methods, 2014, https://doi.org/10.1038/nmeth.2975
Wiltschko et al., Neuron, 2015, https://doi.org/10.1016/j.neuron.2015.11.031
Berman et al., PNAS, 2016, https://doi.org/10.1073/pnas.1607601113
Baldassano et al., Neuron, 2017, https://doi.org/10.1016/j.neuron.2017.06.041
Vidaurre et al., PNAS, 2018, https://doi.org/10.1073/pnas.1705120115
Calhoun, Pillow, Murthy 2019, https://www.nature.com/articles/s41593-019-0533-x
Wiltschko et al., Nature Neuroscience, 2020, https://doi.org/10.1038/s41593-020-00706-3
Recanatesi et al., Neuron, 2022, https://doi.org/10.1016/j.neuron.2021.10.011
Markowitz et al., Nature, 2023, https://doi.org/10.1038/s41586-022-05611-2

Statistical models

Probabilistic models of a dataset $\mathcal{D}$ i.e.: $\mathcal{D} \sim P(\mathcal{D}|\ldots)$

(Random processes perspective; see Lec 2: Fundamentals)

Common uses

Inference of hidden variables

Learning of hidden parameters

Predicting unseen/future observations

Common conventions

$\mathcal{D} \sim P(\mathcal{D}|\Theta)$ Data distribution specified by parameters Θ

$\mathcal{D} \sim P(\mathcal{D}|\mathbf{x}, \Theta)$ Data distribution specified by inputs **x** and parameters Θ

$\mathcal{D} \sim P(\mathcal{D}|\mathbf{x}, \Theta, \mathcal{H})$ Data distribution specified by inputs **x**, parameters Θ, and hyperparameters $\mathcal{H}$

Important time-series models for neuroscience

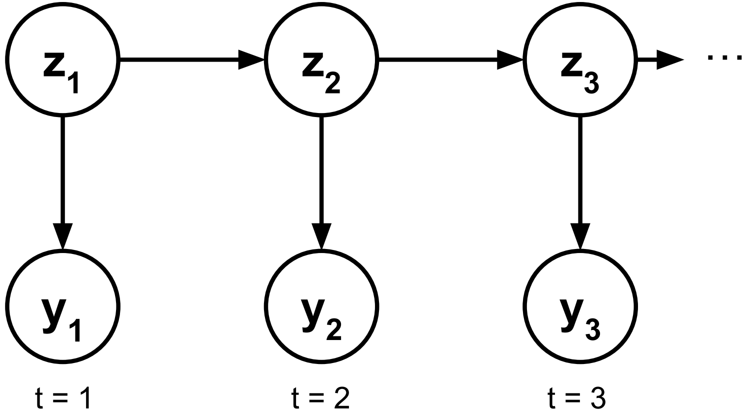
Homogeneous Poisson process: Standard model of spike times given constant firing rate
Inhomogeneous Poisson process: Standard model of spike times given dynamic firing rate
Linear-nonlinear Poisson (LNP) model: Standard model of stimulus-driven spike train
Generalized linear model (GLM): Slightly more general than LNP, since can produce e.g. continuous outputs
Markov Chain (MC): Memoryless model of probabilistic switching among discrete states
Hidden Markov Model (HMM): Hidden variables follow Markov chain, observations are conditioned on hidden variables
Kalman Filter: A continuous version of an HMM, often used for tracking and inference
Variations on Hidden Markov Models: E.g. observations conditioned on hidden variables AND past observations

Directed graphical models

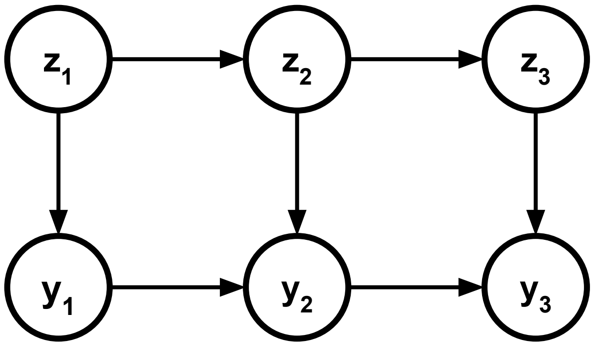
Pictorial summary of joint probability distribution underlying statistical model.
Very common way to illustrate probability model.
For DAG (directed acyclic graphs), specifies how to sample from the distribution.
NOT the same as computation graphs, although can be related.

Common convention:
$\mathbf{z}_t$ → discrete latent
$\mathbf{x}_t$ → continuous latent
$\mathbf{y}_t$ → observed

Example (Hidden Markov Model)



Another example



See Bishop 2006, *Pattern Recognition and Machine Learning*, Ch 8: Graphical Models for an in depth review

Latent variable models

Discrete: $\{\mathbf{y}\} \sim (\{\mathbf{y}\} | \{\mathbf{z}\}, \Theta, \mathcal{H})$

Example: hidden state **z** represents discrete animal state (engaged vs not engaged)

Continuous: $\{\mathbf{y}\} \sim (\{\mathbf{y}\} | \{\mathbf{x}\}, \Theta, \mathcal{H})$

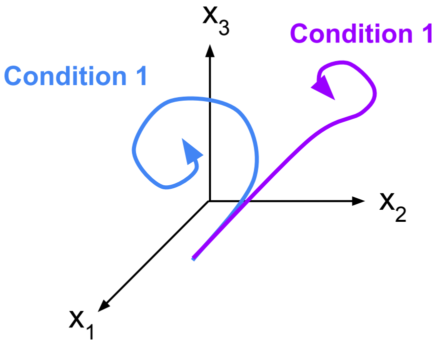
Example: hidden state **x** represents continuous animal state (e.g. arousal level)

Example papers

Smith & Brown, Neural Computation, 2003, https://doi.org/10.1162/089976603765202622
Yu et al., Journal of Neurophysiology, 2009, https://doi.org/10.1152/jn.90941.2008
Pillow et al., Neural Computation, 2010, https://doi.org/10.1162/NECO_a_00058
Fox et al., Annals of Applied Statistics, 2011, https://doi.org/10.1214/10-AOAS395
Linderman et al., ICML, 2017, https://doi.org/10.48550/arXiv.1610.08466
Pandarinath et al., Nature Methods, 2018, https://doi.org/10.1038/s41592-018-0109-9
Ashwood et al 2022 https://www.nature.com/articles/s41593-021-01007-z

State space and latent dynamical systems models

- Enables modeling of time evolution of hidden states underlying observed data.
- Different from imposing statistical model (e.g. Gaussian process) over hidden state time evolution, since state space/LDS usually enforces sequential ("causal") generation of next time points



Latent state **x**(t) evolves via dynamical system

$$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}, \mathbf{u})$$

External inputs

Observations **y**(t), e.g. neural activity, sampled given **x**

$$\mathbf{y} \sim P(\mathbf{y}|\mathbf{x})$$

- In most studies **x** is lower-dimensional than **y**.
- Many different options to choose for f
  - Linear: well understood but not very expressive
  - RNN: More flexible but harder to understand
  - Piecewise-linear: Middle ground between above
  - Represent f as a neural network ("neural ODE")
- One can also model neural activity directly via dynamical system, rather than via latent dynamics (Rajan et al Neuron 2016)

<u>Papers</u>

<u>Linderman et al, *AISTATS* 2017</u>
<u>Pandarinath et al, *Nat Methods* 2018</u>
<u>Luo et al, *Nature* 2025</u>

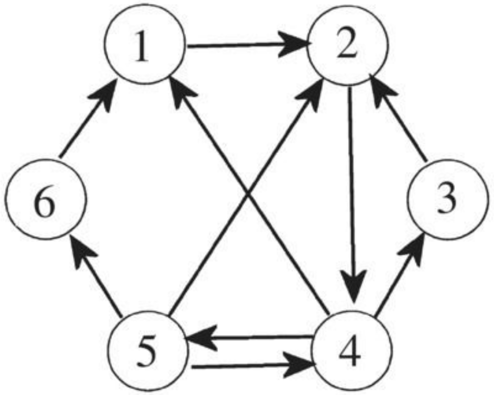
Mechanistic models

Inferring connectivity

Typically infer “functional connectivity”

Many different definitions of functional connectivity

- Correlations between signals from different brain
    - Simple, but no “directionality”
- Granger causality
- Mutual information
- Transfer entropy
- Linear dynamical system connectivity
- Instrumental variable approach



Spatiotemporal analyses

Artificial neural networks

Variational inference & ELBO

Technique for computing posterior over latent variable **z** given data (observations) **x**.

Key idea: $p(\mathbf{z}|\mathbf{x})$ is hard to compute so approximate with $q_\theta(\mathbf{z})$, where q is from family of tractable distributions parameterized by θ.

$p(\mathbf{z})$: prior over latents (easy to compute given **z**)

$p(\mathbf{x}|\mathbf{z})$: likelihood (easy to compute given **x** and **z**)

$p(\mathbf{x},\mathbf{z}) = p(\mathbf{x}|\mathbf{z})p(\mathbf{z})$: joint (easy to compute given **x** and **z**)

Bayes' Rule

$$p(\mathbf{z}|\mathbf{x}) = \frac{p(\mathbf{z})p(\mathbf{x}|\mathbf{z})}{p(\mathbf{x})}$$

Hard to compute

Given data **x** and family of functions $q_\theta(\mathbf{z})$, find θ that minimizes $DKL[q_\theta(\mathbf{z}) \,||\, p(\mathbf{z}|\mathbf{x})]$.

Hard to compute

$$DKL[q_\theta(\mathbf{z}) \,||\, p(\mathbf{z}|\mathbf{x})] = \underbrace{E_q[\log q_\theta(\mathbf{z})] - E_q[p(\mathbf{x},\mathbf{z})]}_{-ELBO(\mathbf{z})} + E_q[\log p(\mathbf{x})]$$

To minimize DKL, maximize the ELBO: $E_q[p(\mathbf{x},\mathbf{z})] - E_q[\log q_\theta(\mathbf{z})]$

Blei et al 2016. Variational Inference: A review for statisticians.

Variational autoencoders

Methods survey - Modern methods - Popular learning/inference techniques

Empirical Bayes

Bayes

Gradient descent

Expectation maximization

Simulation-based inference

Laplace approximation

Forward/backward
algorithms

Methods survey - Modern methods - Information theoretic methods

Max-ent/max-caliber

Schneidman et al *Nature* 2006
Meshulam et al *Neuron* 2017

Mutual info/transfer entropy

Aguera y Arcas, Fairhall, Bialek *Neur Comput* 2003
Kirst, Timme, Battaglia *Nat Commun* 2016
Fairhall, Shea-Brown, Barreiro *Curr Opin Neurobiol* 2018

Predictive information

Bialek, Nemenman, Tishby *Neur Comput* 1999
Palmer et al *PNAS* 2015

Maximally informative dimensions

Sharpee, Rust, Bialek 2003